

Complete Worked PCA Example

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Contents

1	Dataset (Given)	2
2	Step 1 — Feature Means	2
3	Step 2 — Centralized Data (mean-subtraction)	2
4	Step 3 — Covariance Matrix (all workings)	3
5	Step 4 — Eigenvalues (full derivation)	4
6	Step 5 — Eigenvectors (matrix method, full workings for λ_1)	5
7	Step 6 — Projection: Scores (all workings for sample s_1)	8
8	Step 7 — Full Reconstruction (all PCs) — explicit arithmetic	8
9	Step 8 — Reduced Reconstruction (using first 2 PCs), arithmetic	9
10	Step 9 — Additional checks and remarks	10
11	Appendix: Compact numeric matrices used	10

1 Dataset (Given)

We use the small sample dataset with 5 observations (rows) and 3 features (columns).

$$X = \begin{bmatrix} 2 & 0 & -1 \\ 3 & 1 & 0 \\ 2 & 2 & 1 \\ 4 & 0 & 2 \\ 1 & 1 & 1 \end{bmatrix}.$$

Label rows as $s_1, \dots, s_5$ and features as F_1, F_2, F_3 .

2 Step 1 — Feature Means

Compute the mean of each column (feature).

$$\bar{F}_1 = \frac{2 + 3 + 2 + 4 + 1}{5} = \frac{12}{5} = 2.4, \quad \bar{F}_2 = \frac{0 + 1 + 2 + 0 + 1}{5} = \frac{4}{5} = 0.8, \quad \bar{F}_3 = \frac{-1 + 0 + 1 + 2 + 1}{5} = \frac{3}{5}$$

Collect as the mean vector:

$$\bar{\mathbf{x}} = \begin{bmatrix} 2.4 \\ 0.8 \\ 0.6 \end{bmatrix}.$$

3 Step 2 — Centralized Data (mean-subtraction)

Subtract the column means from each row (sample):

For each sample s_i compute $s_{i,c} = s_i - \bar{\mathbf{x}}^\top$.

We show each row explicitly:

$$\begin{aligned} s_{1,c} &= [2, 0, -1] - [2.4, 0.8, 0.6] = [-0.4, -0.8, -1.6], \\ s_{2,c} &= [3, 1, 0] - [2.4, 0.8, 0.6] = [0.6, 0.2, -0.6], \\ s_{3,c} &= [2, 2, 1] - [2.4, 0.8, 0.6] = [-0.4, 1.2, 0.4], \\ s_{4,c} &= [4, 0, 2] - [2.4, 0.8, 0.6] = [1.6, -0.8, 1.4], \\ s_{5,c} &= [1, 1, 1] - [2.4, 0.8, 0.6] = [-1.4, 0.2, 0.4]. \end{aligned}$$

Arrange these rows to form the centralized data matrix X_c :

$$X_c = \begin{bmatrix} -0.4 & -0.8 & -1.6 \\ 0.6 & 0.2 & -0.6 \\ -0.4 & 1.2 & 0.4 \\ 1.6 & -0.8 & 1.4 \\ -1.4 & 0.2 & 0.4 \end{bmatrix}.$$

(Quick check: each column of X_c sums to zero: $-0.4 + 0.6 - 0.4 + 1.6 - 1.4 = 0$, etc.)

4 Step 3 — Covariance Matrix (all workings)

We use the unbiased sample covariance:

$$\text{Cov}(F_i, F_j) = \frac{1}{n-1} \sum_{k=1}^n (x_{k,i} - \bar{F}_i)(x_{k,j} - \bar{F}_j), \quad n = 5 \Rightarrow n-1 = 4.$$

Compute the entries one by one showing sums.

Variances (diagonal entries)

$$\begin{aligned} \sum_{k=1}^5 (F_1 - \bar{F}_1)^2 &= (-0.4)^2 + (0.6)^2 + (-0.4)^2 + (1.6)^2 + (-1.4)^2 \\ &= 0.16 + 0.36 + 0.16 + 2.56 + 1.96 = 5.20. \end{aligned}$$

So

$$\text{Var}(F_1) = \frac{5.20}{4} = 1.30.$$

$$\begin{aligned} \sum_{k=1}^5 (F_2 - \bar{F}_2)^2 &= (-0.8)^2 + (0.2)^2 + (1.2)^2 + (-0.8)^2 + (0.2)^2 \\ &= 0.64 + 0.04 + 1.44 + 0.64 + 0.04 = 2.80. \end{aligned}$$

So

$$\text{Var}(F_2) = \frac{2.80}{4} = 0.70.$$

$$\begin{aligned} \sum_{k=1}^5 (F_3 - \bar{F}_3)^2 &= (-1.6)^2 + (-0.6)^2 + (0.4)^2 + (1.4)^2 + (0.4)^2 \\ &= 2.56 + 0.36 + 0.16 + 1.96 + 0.16 = 5.20. \end{aligned}$$

So

$$\text{Var}(F_3) = \frac{5.20}{4} = 1.30.$$

Covariances (off-diagonal)

$$\begin{aligned} \sum_{k=1}^5 (F_1 - \bar{F}_1)(F_2 - \bar{F}_2) &= (-0.4)(-0.8) + (0.6)(0.2) + (-0.4)(1.2) + (1.6)(-0.8) + (-1.4)(0.2) \\ &= 0.32 + 0.12 - 0.48 - 1.28 - 0.28 = -1.60. \end{aligned}$$

So

$$\text{Cov}(F_1, F_2) = \frac{-1.60}{4} = -0.40.$$

$$\begin{aligned} \sum_{k=1}^5 (F_1 - \bar{F}_1)(F_3 - \bar{F}_3) &= (-0.4)(-1.6) + (0.6)(-0.6) + (-0.4)(0.4) + (1.6)(1.4) + (-1.4)(0.4) \\ &= 0.64 - 0.36 - 0.16 + 2.24 - 0.56 = 1.80. \end{aligned}$$

So

$$\text{Cov}(F_1, F_3) = \frac{1.80}{4} = 0.45.$$

$$\begin{aligned}\sum_{k=1}^5 (F_2 - \bar{F}_2)(F_3 - \bar{F}_3) &= (-0.8)(-1.6) + (0.2)(-0.6) + (1.2)(0.4) + (-0.8)(1.4) + (0.2)(0.4) \\ &= 1.28 - 0.12 + 0.48 - 1.12 + 0.08 = 0.60.\end{aligned}$$

So

$$\text{Cov}(F_2, F_3) = \frac{0.60}{4} = 0.15.$$

Covariance matrix

Collecting the values (symmetry enforced):

$$\Sigma = \text{Cov}(X) = \begin{bmatrix} 1.30 & -0.40 & 0.45 \\ -0.40 & 0.70 & 0.15 \\ 0.45 & 0.15 & 1.30 \end{bmatrix}.$$

(Quick checks: trace = $1.30 + 0.70 + 1.30 = 3.30$.)

5 Step 4 — Eigenvalues (full derivation)

We solve the characteristic equation:

$$\det(\Sigma - \lambda I) = 0.$$

Write the matrix:

$$\Sigma - \lambda I = \begin{bmatrix} 1.30 - \lambda & -0.40 & 0.45 \\ -0.40 & 0.70 - \lambda & 0.15 \\ 0.45 & 0.15 & 1.30 - \lambda \end{bmatrix}.$$

To reduce algebraic error, convert to rational fractions where convenient:

$$1.30 = \frac{13}{10}, \quad 0.70 = \frac{7}{10}, \quad -0.40 = -\frac{2}{5}, \quad 0.45 = \frac{9}{20}, \quad 0.15 = \frac{3}{20}.$$

Compute determinant by expansion (step-by-step)

Expand along the first row:

$$\det(\Sigma - \lambda I) = \left(\frac{13}{10} - \lambda\right) \begin{vmatrix} \frac{7}{10} - \lambda & \frac{3}{20} \\ \frac{3}{20} & \frac{13}{10} - \lambda \end{vmatrix} - \left(-\frac{2}{5}\right) \begin{vmatrix} -\frac{2}{5} & \frac{3}{20} \\ \frac{9}{20} & \frac{13}{10} - \lambda \end{vmatrix} + \left(\frac{9}{20}\right) \begin{vmatrix} -\frac{2}{5} & \frac{7}{10} - \lambda \\ \frac{9}{20} & \frac{3}{20} \end{vmatrix}.$$

Now compute each 2×2 minor carefully.

First minor

$$\begin{aligned}M_1 &= \left(\frac{7}{10} - \lambda\right) \left(\frac{13}{10} - \lambda\right) - \left(\frac{3}{20}\right)^2 \\ &= \lambda^2 - \left(\frac{7}{10} + \frac{13}{10}\right)\lambda + \frac{7}{10} \cdot \frac{13}{10} - \frac{9}{400} \\ &= \lambda^2 - 2\lambda + \left(\frac{91}{100} - \frac{9}{400}\right) = \lambda^2 - 2\lambda + \frac{355}{400}.\end{aligned}$$

Second minor

$$\begin{aligned}
M_2 &= -\frac{2}{5} \left(\frac{13}{10} - \lambda \right) - \frac{3}{20} \cdot \frac{9}{20} \\
&= -\frac{26}{50} + \frac{2}{5} \lambda - \frac{27}{400} = \frac{2}{5} \lambda - \frac{235}{400}.
\end{aligned}$$

Third minor

$$\begin{aligned}
M_3 &= -\frac{2}{5} \cdot \frac{3}{20} - \frac{9}{20} \left(\frac{7}{10} - \lambda \right) \\
&= -\frac{6}{100} - \frac{63}{200} + \frac{9}{20} \lambda = \frac{9}{20} \lambda - \frac{75}{200} = \frac{9}{20} \lambda - \frac{3}{8}.
\end{aligned}$$

Plug back and expand

$$\det(\Sigma - \lambda I) = \left(\frac{13}{10} - \lambda \right) \left(\lambda^2 - 2\lambda + \frac{355}{400} \right) + \frac{2}{5} \left(\frac{2}{5} \lambda - \frac{235}{400} \right) + \frac{9}{20} \left(\frac{9}{20} \lambda - \frac{3}{8} \right).$$

Carrying out the algebra (combine like terms carefully) yields the cubic polynomial:

$$\lambda^3 - \frac{33}{10} \lambda^2 + \frac{25}{8} \lambda - \frac{3}{4} = 0.$$

(As decimals: $\lambda^3 - 3.3\lambda^2 + 3.125\lambda - 0.75 = 0$.)

Numeric roots (eigenvalues)

Solving that cubic numerically (standard root-finding or numeric diagonalization) gives three real eigenvalues:

$$\lambda_1 \approx 1.7838959164, \quad \lambda_2 \approx 1.1507539411, \quad \lambda_3 \approx 0.3653501426.$$

Sanity checks

- Trace check: $\lambda_1 + \lambda_2 + \lambda_3 \approx 1.783896 + 1.150754 + 0.365350 = 3.300000 \approx \text{tr}(\Sigma)$.
- Determinant check: $\lambda_1 \lambda_2 \lambda_3 \approx 0.75 = \det(\Sigma)$ (matches earlier algebra).
- All eigenvalues are non-negative (covariance matrix positive semi-definite).

6 Step 5 — Eigenvectors (matrix method, full workings for λ_1)

Method reminder

For each eigenvalue λ solve:

$$(\Sigma - \lambda I)\mathbf{v} = \mathbf{0}.$$

We perform Gaussian elimination to find the nullspace. After finding a nonzero solution $\mathbf{v}$, we normalize to unit length to obtain $\mathbf{u}$.

Compute $\Sigma - \lambda_1 I$ (numeric approximate)

Using $\lambda_1 \approx 1.7838959164$,

$$\Sigma - \lambda_1 I \approx \begin{bmatrix} 1.30 - \lambda_1 & -0.40 & 0.45 \\ -0.40 & 0.70 - \lambda_1 & 0.15 \\ 0.45 & 0.15 & 1.30 - \lambda_1 \end{bmatrix} = \begin{bmatrix} -0.4838959164 & -0.4000000000 & 0.4500000000 \\ -0.4000000000 & -1.0838959164 & 0.1500000000 \\ 0.4500000000 & 0.1500000000 & -0.4838959164 \end{bmatrix}.$$

Solve

$$\begin{bmatrix} -0.4838959164 & -0.4 & 0.45 \\ -0.4 & -1.0838959164 & 0.15 \\ 0.45 & 0.15 & -0.4838959164 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Gaussian elimination — full numeric steps

We show row operations with numeric values (rounded to 9 d.p. where listed).

Start with augmented matrix (no RHS shown since it's zero):

$$\begin{bmatrix} -0.4838959164 & -0.4000000000 & 0.4500000000 \\ -0.4000000000 & -1.0838959164 & 0.1500000000 \\ 0.4500000000 & 0.1500000000 & -0.4838959164 \end{bmatrix}.$$

Step 1 — make leading 1 in row 1: Divide row1 by -0.4838959164 :

$$R_1 \leftarrow R_1 / (-0.4838959164) \approx [1 \quad 0.8266239151 \quad -0.9299518897].$$

Step 2 — eliminate first column in rows 2 and 3 Eliminate using R_1 :

$$- R_2 \leftarrow R_2 - (-0.4) \cdot R_1 = R_2 + 0.4R_1:$$

$$\begin{aligned} R_2 &\approx [-0.4, -1.0838959164, 0.15] + 0.4[1, 0.8266239151, -0.9299518897] \\ &= [-0.4 + 0.4, -1.0838959164 + 0.3306495660, 0.15 - 0.3719807559] \\ &\approx [0, -0.7532463504, -0.2219807559]. \end{aligned}$$

$$- R_3 \leftarrow R_3 - 0.45 \cdot R_1:$$

$$\begin{aligned} R_3 &\approx [0.45, 0.15, -0.4838959164] - 0.45[1, 0.8266239151, -0.9299518897] \\ &= [0.45 - 0.45, 0.15 - 0.372, -0.4838959164 + 0.4184783504] \end{aligned}$$

(calculate precisely): $0.15 - 0.372$, where $0.45 \times 0.8266239151 = 0.372$,

$$R_3 \approx [0, -0.222, -0.0654175653] \approx [0, -0.2219807559, -0.0654175653].$$

(Observe R_2 and R_3 now equal to within symmetry; numeric rounding yields:)

$$R_2 \approx [0, -0.7532463504, -0.2219807559], \quad R_3 \approx [0, -0.2219807559, -0.0654175653].$$

Step 3 — normalize row 2 Divide R_2 by its leading value -0.7532463504 :

$$R_2 \leftarrow R_2 / (-0.7532463504) \approx [0, 1, 0.2946987097].$$

Step 4 — eliminate second column in row 3 Eliminate with R_2 :

$$R_3 \leftarrow R_3 - (-0.2219807559) \cdot R_2 = R_3 + 0.2219807559 \cdot R_2.$$

Compute:

$$R_3 \approx [0, -0.2219807559, -0.0654175653] + 0.2219807559[0, 1, 0.2946987097] = [0, 0, 0]$$

(with numerical rounding the third entry becomes order 10^{-16} , effectively zero).

Conclusion of elimination We obtain row-echelon form:

$$\begin{bmatrix} 1 & 0.8266239151 & -0.9299518897 \\ 0 & 1 & 0.2946987097 \\ 0 & 0 & 0 \end{bmatrix}.$$

Back-substitution to express variables

From row 2:

$$y + 0.2946987097z = 0 \quad \Rightarrow \quad y = -0.2946987097z.$$

From row 1:

$$x + 0.8266239151y - 0.9299518897z = 0.$$

Substitute y :

$$x + 0.8266239151(-0.2946987097z) - 0.9299518897z = 0.$$

Compute coefficient:

$$0.8266239151 \times (-0.2946987097) \approx -0.2436050653.$$

Thus:

$$x - 0.2436050653z - 0.9299518897z = 0 \quad \Rightarrow \quad x = (0.2436050653 + 0.9299518897)z.$$

Sum:

$$x \approx 1.1735569550z.$$

Choose scale Pick $z = 1$ so a particular eigenvector is

$$\mathbf{v}^{(1)} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \approx \begin{bmatrix} 1.1735569550 \\ -0.2946987097 \\ 1 \end{bmatrix}.$$

Normalization

Compute norm:

$$\|\mathbf{v}^{(1)}\| = \sqrt{1.1735569550^2 + (-0.2946987097)^2 + 1^2} \approx \sqrt{1.377243 + 0.086836 + 1} \approx \sqrt{2.464079} \approx 1.570243.$$

Normalize:

$$\mathbf{u}_1 = \frac{\mathbf{v}^{(1)}}{\|\mathbf{v}^{(1)}\|} \approx \begin{bmatrix} 1.1735569550/1.570243 \\ -0.2946987097/1.570243 \\ 1/1.570243 \end{bmatrix} \approx \begin{bmatrix} 0.74761244 \\ -0.18773736 \\ 0.63704814 \end{bmatrix}.$$

Eigenvectors for λ_2 and λ_3 (condensed workings)

We apply the same procedure to $\lambda_2 \approx 1.1507539411$ and $\lambda_3 \approx 0.3653501426$. For brevity we show the core steps and the resulting normalized vectors.

For λ_2 : Form $(\Sigma - \lambda_2 I)$ and perform Gaussian elimination (numeric). After back-substitution and normalizing we obtain:

$$\mathbf{u}_2 \approx \begin{bmatrix} 0.42488991 \\ -0.60202040 \\ -0.67604734 \end{bmatrix}.$$

For λ_3 : Similarly solving for λ_3 yields:

$$\mathbf{u}_3 \approx \begin{bmatrix} 0.51043531 \\ 0.77609673 \\ -0.37031023 \end{bmatrix}.$$

Assemble orthonormal matrix

Put eigenvectors as columns (aligned with eigenvalues $\lambda_1, \lambda_2, \lambda_3$):

$$U = \begin{bmatrix} 0.74761244 & 0.42488991 & 0.51043531 \\ -0.18773736 & -0.60202040 & 0.77609673 \\ 0.63704814 & -0.67604734 & -0.37031023 \end{bmatrix}.$$

Orthogonality check (numeric) Compute $U^\top U \approx I$ (small off-diagonal entries on order 10^{-9} due to rounding).

7 Step 6 — Projection: Scores (all workings for sample s_1)

Given centralized sample $s_{1,c} = [-0.4, -0.8, -1.6]^\top$, we compute the PC scores:

$$\mathbf{y} = U^\top s_{1,c} \quad \text{or} \quad \mathbf{y} = s_{1,c}^\top U.$$

We compute each coordinate as a dot product.

PC1 score

$$\begin{aligned} y_1 &= s_{1,c} \cdot \mathbf{u}_1 \\ &= (-0.4)(0.74761244) + (-0.8)(-0.18773736) + (-1.6)(0.63704814) \\ &= -0.299044976 + 0.150189888 - 1.019277024 \\ &= -1.168132112 \approx -1.16813211. \end{aligned}$$

PC2 score

$$\begin{aligned} y_2 &= (-0.4)(0.42488991) + (-0.8)(-0.60202040) + (-1.6)(-0.67604734) \\ &= -0.169955964 + 0.481616320 + 1.081675744 \\ &= 1.393336100 \approx 1.39333610. \end{aligned}$$

PC3 score

$$\begin{aligned} y_3 &= (-0.4)(0.51043531) + (-0.8)(0.77609673) + (-1.6)(-0.37031023) \\ &= -0.204174124 - 0.620877384 + 0.592496368 \\ &= -0.232555140 \approx -0.23255513. \end{aligned}$$

Collect:

$$\mathbf{y} \approx \begin{bmatrix} -1.16813211 \\ 1.39333610 \\ -0.23255513 \end{bmatrix}.$$

8 Step 7 — Full Reconstruction (all PCs) — explicit arithmetic

Reconstruction formula (mean added back):

$$\hat{s}_1 = U\mathbf{y} + \bar{\mathbf{x}}.$$

We compute each contribution $y_j \mathbf{u}_j$ explicitly.

Contribution from PC1

$$\begin{aligned} y_1 \mathbf{u}_1 &= -1.16813211 \begin{bmatrix} 0.74761244 \\ -0.18773736 \\ 0.63704814 \end{bmatrix} = \begin{bmatrix} -1.16813211 \times 0.74761244 \\ -1.16813211 \times (-0.18773736) \\ -1.16813211 \times 0.63704814 \end{bmatrix} \\ &\approx \begin{bmatrix} -0.873310100 \\ 0.219302036 \\ -0.744156136 \end{bmatrix}. \end{aligned}$$

Contribution from PC2

$$y_2 \mathbf{u}_2 = 1.39333610 \begin{bmatrix} 0.42488991 \\ -0.60202040 \\ -0.67604734 \end{bmatrix} \approx \begin{bmatrix} 0.592014450 \\ -0.838816760 \\ -0.941961160 \end{bmatrix}.$$

Contribution from PC3

$$y_3 \mathbf{u}_3 = -0.23255513 \begin{bmatrix} 0.51043531 \\ 0.77609673 \\ -0.37031023 \end{bmatrix} \approx \begin{bmatrix} -0.118704350 \\ -0.180485280 \\ 0.086117552 \end{bmatrix}.$$

Sum of contributions = reconstructed centralized vector

$$\begin{aligned} \hat{s}_{1,c} &= y_1 \mathbf{u}_1 + y_2 \mathbf{u}_2 + y_3 \mathbf{u}_3 \\ &\approx \begin{bmatrix} -0.873310100 \\ 0.219302036 \\ -0.744156136 \end{bmatrix} + \begin{bmatrix} 0.592014450 \\ -0.838816760 \\ -0.941961160 \end{bmatrix} + \begin{bmatrix} -0.118704350 \\ -0.180485280 \\ 0.086117552 \end{bmatrix} \\ &= \begin{bmatrix} -0.873310100 + 0.592014450 - 0.118704350 \\ 0.219302036 - 0.838816760 - 0.180485280 \\ -0.744156136 - 0.941961160 + 0.086117552 \end{bmatrix} = \begin{bmatrix} -0.4 \\ -0.8 \\ -1.6 \end{bmatrix}. \end{aligned}$$

Add back mean

$$\hat{s}_1 = \hat{s}_{1,c} + \bar{\mathbf{x}} = \begin{bmatrix} -0.4 \\ -0.8 \\ -1.6 \end{bmatrix} + \begin{bmatrix} 2.4 \\ 0.8 \\ 0.6 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix},$$

which exactly recovers the original sample (up to floating-point tiny noise).

9 Step 8 — Reduced Reconstruction (using first 2 PCs), arithmetic

Set $y_3 = 0$ and reconstruct from y_1, y_2 only.

Sum of PC1 and PC2 contributions

$$\begin{aligned} \hat{s}_{1,c}^{(2)} &= y_1 \mathbf{u}_1 + y_2 \mathbf{u}_2 \\ &\approx \begin{bmatrix} -0.873310100 \\ 0.219302036 \\ -0.744156136 \end{bmatrix} + \begin{bmatrix} 0.592014450 \\ -0.838816760 \\ -0.941961160 \end{bmatrix} = \begin{bmatrix} -0.281295650 \\ -0.619514724 \\ -1.686117296 \end{bmatrix}. \end{aligned}$$

(Add small rounding corrections to match earlier slightly different decimals.)

Add mean

$$\hat{s}_1^{(2)} = \hat{s}_{1,c}^{(2)} + \bar{\mathbf{x}} \approx \begin{bmatrix} 2.11870435 \\ 0.18048528 \\ -1.08611730 \end{bmatrix}.$$

Reconstruction error (sample-wise)

$$\mathbf{e} = s_1 - \hat{s}_1^{(2)} \approx \begin{bmatrix} -0.11870435 \\ -0.18048528 \\ 0.08611730 \end{bmatrix}.$$

Squared error sum $\approx 0.1187^2 + 0.1805^2 + 0.0861^2 \approx 0.035656$ (contributes to the overall MSE when reduced to 2 PCs).

10 Step 9 — Additional checks and remarks

- **Total variance** = $\text{trace}(\Sigma) = 3.3$. Sum of eigenvalues = $1.7839 + 1.1508 + 0.36535 \approx 3.3$ (matches).
- **Variance explained:**

$$\frac{\lambda_1}{\sum \lambda_i} \approx \frac{1.7839}{3.3} \approx 54.06\%, \quad \frac{\lambda_2}{3.3} \approx 34.87\%, \quad \frac{\lambda_3}{3.3} \approx 11.07\%.$$

- **Full reconstruction with all PCs** recovers the original exactly (because we used an orthonormal basis spanning the centered data space).
- **Reduced reconstruction** discards variance equal to the sum of omitted eigenvalues (here omitting PC3 discards ≈ 0.36535 variance).

11 Appendix: Compact numeric matrices used

For reference, the main numeric matrices used (rounded to 8 dp) are:

$$X_c = \begin{bmatrix} -0.4 & -0.8 & -1.6 \\ 0.6 & 0.2 & -0.6 \\ -0.4 & 1.2 & 0.4 \\ 1.6 & -0.8 & 1.4 \\ -1.4 & 0.2 & 0.4 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 1.30 & -0.40 & 0.45 \\ -0.40 & 0.70 & 0.15 \\ 0.45 & 0.15 & 1.30 \end{bmatrix},$$

$$U = \begin{bmatrix} 0.74761244 & 0.42488991 & 0.51043531 \\ -0.18773736 & -0.60202040 & 0.77609673 \\ 0.63704814 & -0.67604734 & -0.37031023 \end{bmatrix}, \quad \Lambda = \text{diag}(1.78389592, 1.15075394, 0.36535014).$$